Traffic Congestion and Urban Heat: Evidence from Freeway Incidents in California

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October 20, 2026

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Abstract

Traffic congestion is a plausible contributor to localized urban heat, a costly externality for the many who live and work near major roads. Yet its short-run causal effect on local surface temperature has not been credibly identified: existing estimates are associational or rely on broad natural experiments. We estimate it for California freeways using unplanned accidents as an instrument for congestion. The central threat is reverse causality: because congestion raises accident risk, accidents are in general endogenous to the congestion they are meant to instrument. We document this bias with an event study and remove it by restricting the instrument to accidents that strike while the segment is free-flowing, before congestion has formed. We measure land surface temperature from NASA ECOSTRESS at 70-meter resolution and exploit its varying overpass time to match each reading to same-hour congestion. Our central estimate is the reduced form: a free-flow-onset accident raises near-road surface temperature by about 0.11°C over the following two hours, against flat pre-trends. The increase is about half of the 0.21°C by which the near-road ring is on average warmer than the 500–1,000-meter outer ring. Scaled by the accident's effect on congestion, the reduced form implies roughly 0.6 to 1.1°C per 10 mph of speed reduction. The response is largest in the near-road ring and declines significantly with distance, tracing how a localized transport externality fades into the surrounding area.

Keywords: urban heat island, traffic congestion, land surface temperature, instrumental variables, freeway incidents, ECOSTRESS, California

JEL Classification: Q54, R41, C26

*Assistant Professor of Economics, Xavier University. Contact: arnolds7@xavier.edu. I thank participants at the ASHEcon Annual Conference and at the Miami University brown bag seminar for helpful comments and suggestions. This study contains modified Copernicus Climate Change Service information (2018–2026); neither the European Commission nor ECMWF is responsible for any use that may be made of the Copernicus information or data it contains. Any errors are my own.

1 Introduction

How far from the road does traffic congestion raise temperature? Extreme heat is the leading cause of weather-related deaths in the United States, exceeding floods, tornadoes, and hurricanes in average annual fatalities (National Weather Service, 2024). Urban heat, the systematic warmth of developed areas relative to their surroundings, carries well-documented health and energy costs (Heaviside et al., 2017; Santamouris et al., 2015). Those costs are concentrated on the hottest days: mortality is flat over a wide range of daily temperatures and rises steeply at the hot end (Deschênes and Greenstone, 2011; Gasparrini et al., 2015; Carleton et al., 2022), so an increment of a given size does most of its damage on a day that is already hot. Its structural drivers, the pavement, rooftops, and sparse vegetation that set a neighborhood's baseline temperature, are extensively studied (Oke, 1973; Gago et al., 2013). Much less is known about its time-varying drivers, and in particular about traffic congestion, which fluctuates hour to hour over a fixed road network that millions of people live beside. About 11 million Californians reside within one kilometer of the freeway detectors we study, 4.7 million of them within 500 meters. Whether congestion measurably raises temperature around these freeways, and how far from the road the effect extends, is an understudied spatial question.

It is also an urban question. Climate is a priced amenity: measured through wages and rents, American households pay more on the margin to avoid a degree of excess heat than a degree of excess cold (Albouy et al., 2016). A localized disamenity is likewise capitalized into housing values within about a kilometer of its source (Currie et al., 2015). Indoor temperatures in city homes follow outdoor heat with a lag of a few hours and differ widely across the housing stock (White-Newsome et al., 2012; Quinn et al., 2014). In the 2003 Paris heat wave, the risk of death rose with the nighttime surface temperature around a resident's home (Laaidi et al., 2012). The exposure is unequal: in nearly every large United States urbanized area, people of color and households in poverty live in hotter census tracts (Hsu et al., 2021), and formerly redlined neighborhoods are the hotter ones in 94 percent of 108

urban areas (Hoffman et al., 2020). Beyond mortality, short-run heat reduces labor supply in exposed industries (Graff Zivin and Neidell, 2014), slows learning (Park et al., 2020), and raises violent crime, most in poor neighborhoods (Heilmann et al., 2021).

The mechanism is well grounded: vehicles in stop-and-go operation burn more fuel per mile than vehicles at cruise speed (Barth and Boriboonsomsin, 2008), and so release more waste heat. The anthropogenic heat-flux literature finds traffic to be a substantial, commute-peaked share of the heat that human activity releases in cities (Sailor and Lu, 2004; Sailor, 2011; Quah and Roth, 2012). But causal identification of the congestion-to-heat link is limited. The closest evidence (Lee and Berkelhammer, 2025) relates bus speeds to satellite surface temperature in Chicago, an association rather than a causal estimate. Quasi-experimental alternatives rely on shocks, such as highway expansions (Yang et al., 2026) and COVID-19 mobility restrictions (Tran et al., 2026), that move land use, economic activity, and human presence alongside traffic.

We estimate the effect of freeway congestion on nearby surface temperature using unplanned freeway accidents logged in the Caltrans Performance Measurement System (PeMS) as an instrument for congestion. The central threat to this design is reverse causality: congestion itself raises accident risk. We therefore restrict the instrument to free-flow-onset accidents, those that strike while the segment is still free-flowing, before any congestion has formed. An event study validates the restriction: the speed performance index (realized speed relative to free-flow speed) shows no congestion building before a free-flow-onset accident and falls when the accident occurs. Surface-temperature pre-trends are flat (Section 5.1). We measure the temperature response with NASA ECOSTRESS, whose 70-meter resolution and varying overpass times let each land surface temperature (LST) reading be matched to a PeMS congestion reading from the same hour. We estimate the response in concentric distance bands around each detector, in the spirit of Currie and Walker (2011).

Our central estimate is the reduced form: a free-flow-onset accident raises near-road surface temperature by about 0.11°C over the following two hours, against flat pre-trends.

Scaled by the accident's effect on congestion, this implies roughly 0.6 to 1.1°C per 10 mph reduction in speed. The effect is positive in every distance ring, largest at the road, and declines significantly from the near-road ring to the outer one. The raw correlation cannot be read as this effect: conditional on detector, date, and hour-of-day fixed effects, the OLS association is small and wrong-signed.

Standard accountings of the external costs of driving count congestion, accidents, and pollution, with noise and pavement damage far behind (Parry and Small, 2005). Heat appears in none of them, so the thermal channel is a local externality that no current policy prices. It is also one that existing policy already moves. Because the estimates are identified from accident-induced congestion episodes, they bear on what rapid incident detection and clearance also buy back in near-road heat. An evaluation of California's Freeway Service Patrol, which made over 686,000 assists in 2017–18, counts its delay, fuel, and emission savings (Mauch et al., 2019).

The paper makes three contributions. First, it provides a causal estimate of the short-run effect of traffic congestion on local surface temperature. Second, it shows that accidents, used as instruments for congestion precisely because they appear exogenous (Beland and Brent, 2018), are systematically endogenous to the congestion they are meant to instrument. It also shows that conditioning on the pre-accident state removes the reverse causality. Third, it measures how the congestion-heat response declines with distance from the road.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature and Section 3 describes the data. Section 4 presents the research design: the event study and the reduced-form and two-stage least squares (2SLS) specifications. Section 5 reports the results: the near-road temperature response, its decline with distance from the road, and its size per unit of congestion. Section 6 reports the robustness checks central to identification, with further checks in the Appendix, and Section 7 concludes.

2 Related Literature

Urban heat islands arise when vegetated, permeable surfaces are replaced with impervious materials that absorb and retain solar radiation while eliminating evaporative cooling (Oke, 1973; Gago et al., 2013; Declet-Barreto et al., 2013). The stakes are well quantified. Heaviside et al. (2016) attribute roughly half of heat-related deaths in the 2003 West Midlands heatwave to the urban heat island. The heat island itself intensifies during heat waves, when low wind and dry surfaces cut the losses that would carry added heat away (Li and Bou-Zeid, 2013; Zhao et al., 2018). Tuholske et al. (2021) document a nearly 200 percent increase in urban population exposure to extreme heat between 1983 and 2016. Santamouris et al. (2015) review studies in which each degree of warming raises peak electricity demand by 0.45 to 4.6 percent. This literature centers on structural features of the built environment that vary slowly over time, measured at scale through thermal infrared remote sensing (Li et al., 2020).

Direct evidence linking congestion to local temperature is scarce. Husni et al. (2022) use IoT sensors and traffic cameras in Indonesia to document air-temperature increases of up to 7°C during severe congestion on sunny days. Yang et al. (2026) find that Bay Area highway expansions exacerbate urban–rural temperature differentials in Moderate Resolution Imaging Spectroradiometer (MODIS) data. Lee and Berkelhammer (2025) estimate a 0.36°C increase in surface urban heat island intensity per 10 mph decline in Chicago bus speeds.

A larger economics literature establishes that air pollution imposes substantial costs on infant health, productivity, and housing values (Graff Zivin and Neidell, 2012; Chay and Greenstone, 2003, 2005), and that traffic is a direct source of it (Knittel et al., 2016; Isaksen and Johansen, 2021). Xia et al. (2023) study the within-city effects of congestion on NO_2 and CO, the closest pollution-side analog to our congestion-temperature design. Currie and Walker (2011) compare birth outcomes for mothers living within two kilometers of a toll plaza with those two to ten kilometers away, before and after the introduction of E-ZPass reduced idling and congestion at the plazas. They find improved outcomes for the nearby mothers. To instrument for congestion itself, rather than infrastructure (where Duranton

and Turner (2011) established the canonical design), prior work has used transit strikes (Anderson, 2014) and other transitory shocks to traffic. Unplanned freeway accidents add a shock located to the detector segment and timestamped to the minute. This approach draws on a broad tradition in economics that exploits the idiosyncratic short-run timing of natural shocks, weather most prominently, as a source of exogenous variation (Miguel et al., 2004; Deschênes and Greenstone, 2007).

Two further literatures bear directly on the design. First, accidents have been used in economics as a source of exogenous variation in traffic. Studying the effect of congestion on crime, Beland and Brent (2018) report a specification that instruments extreme traffic in Los Angeles with Caltrans-logged accidents delaying traffic by more than an hour. The specification draws on the same PeMS detector data we use. Transportation research, however, shows that freeway crash risk rises with congestion and can be predicted from detector readings in the minutes before a crash (Abdel-Aty et al., 2004; Yeo et al., 2013). Accidents therefore cannot be assumed exogenous to the congestion around them. That the instrument appears in published work on our own data source is what makes the endogeneity we document consequential beyond this setting. Second, a body of work identifies near-road exposure from wind direction relative to the source (Anderson, 2020; Herrnstadt et al., 2021), a comparison we also report for heat.

Finally, the paper's spatial object, the distance gradient of the temperature response, connects to a regional-science tradition tracing how localized disamenities decay with distance from their source. Currie et al. (2015) measure the reach of toxic-plant emissions into surrounding housing markets, and Brinkman and Lin (2024) value the local quality-of-life costs of urban highways.

We add a causal estimate of how a time-varying driver of urban heat, freeway congestion, affects surface temperature across California's metropolitan freeways. The estimate is identified from accidents that strike free-flowing segments, with temperature matched to traffic conditions hour by hour through the 70-meter, varying-overpass-time ECOSTRESS

sensor (Fisher et al., 2020). It extends the study of traffic's external costs from pollution, a chemical pathway, to a thermal one. We also measure the distance gradient of the response in physical rather than price terms, asking how far the temperature response to congestion extends into the area around the network.

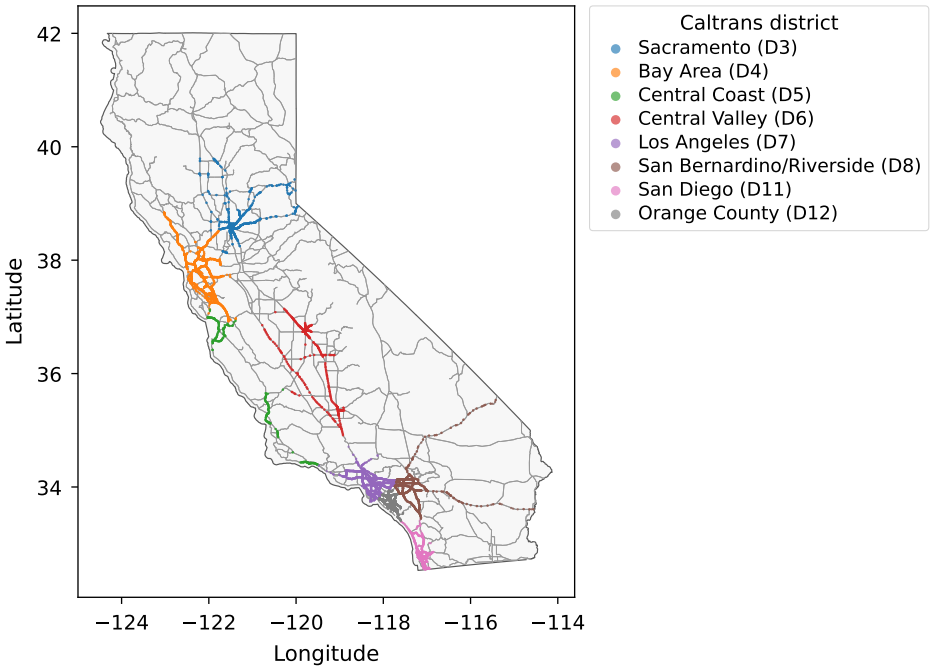
3 Data

3.1 Traffic Congestion and Incident Logs: PeMS

Traffic data come from the Caltrans Performance Measurement System (PeMS; California Department of Transportation, 2026), which operates a network of traffic detectors embedded in California freeway lanes. Each detector records flow (vehicles per hour), occupancy (share of time a vehicle is present over the sensor), and speed (miles per hour) at five-minute intervals. Detectors are spaced approximately every half mile along major California freeways and are organized into stations that aggregate across lanes. We obtain station-level hourly data and station metadata (GPS coordinates, freeway route, and direction of travel) for eight Caltrans districts from June 2018 through March 2026.[1] Figure 1 maps the detectors in the estimation panel (Table 1) across these eight districts.

[1]We include the eight Caltrans districts covering California's major metropolitan areas. They are District 3 (Sacramento), District 4 (Bay Area), District 5 (San Luis Obispo/Central Coast), District 6 (Fresno/Central Valley), District 7 (Los Angeles), District 8 (San Bernardino/Riverside), District 11 (San Diego), and District 12 (Orange County). Districts 1 (North Coast), 2 (Northern California), 9 (Eastern Sierra), and 10 (Central Sierra) are excluded because they cover predominantly rural and mountainous terrain with sparse freeway detector networks and limited urban heat island exposure. For four districts (6, 7, 11, and 12) the most recent station-metadata snapshot predates some newer detectors, which therefore do not enter the panel.

Figure 1: PeMS freeway detectors in the estimation panel, by Caltrans district



The primary congestion measure is the speed performance index (He et al., 2016), which we compute as the ratio of realized speed to free-flow speed,

$$\text{SPI}_{it} = \frac{\text{Speed}_{it}}{\text{Free-Flow Speed}_{i}}$$

where i indexes the PeMS detector station and t indexes the hour of the ECOSTRESS overpass. A value of 1 indicates free-flow conditions and values approaching 0 indicate near-standstill conditions, so the index falls as congestion worsens. Free-flow speed is defined empirically as the 85th percentile of observed speeds at station i during overnight hours (10pm–5am) and weekends, when congestion is minimal. This percentile-based definition makes the index comparable across segments with different posted speed limits and traffic compositions.

For the incident instrument, we use PeMS incident logs, which record the timestamp, segment, type, and duration of each incident logged on California freeways. PeMS classifies incidents into a range of categories, the most common of which are traffic collisions,

traffic hazards, animal hazards, vehicle fires, road and weather conditions, and assists to Caltrans operations. We restrict the instrument to traffic accidents (collisions, hit-and-run collisions, and spinouts), which are the most plausibly exogenous to neighborhood characteristics and environmental conditions. Weather-related incidents are excluded because they reflect weather, which affects surface temperature directly and enters the specification as a control. Traffic hazards, animal hazards, debris, and planned events such as construction assists and road closures are excluded because they either do not reliably generate congestion or are not unplanned. We return to weather and vehicle fires in Section 4.2. Each accident is timestamped when the California Highway Patrol (CHP) logs it and matched to the nearest mainline detector within one kilometer on the same freeway and in the same direction. About 1.2 million accidents are matched over the study period. We classify onset using the speed performance index at that detector over the clock hour preceding the hour in which the accident is logged (the pre-accident hour; Section 4.1). About half of accidents cannot be classified because the detector reports no speed in that hour; these, like congested-onset accidents (those striking already-congested segments), are excluded from the estimation sample rather than counted as controls.

The PeMS incident archive is largely complete over the study period, except for March–early July 2025, when the feed was not reliably archived.[2] We treat the instrument as undefined on dates lacking reliable incident records (missing, empty, or anomalously sparse) and drop ECOSTRESS overpasses falling in those windows. The 2025 gap therefore removes roughly four months of overpasses from the estimation panel. Because the gap is a data-archiving artifact unrelated to traffic, weather, or neighborhood conditions, this attrition is plausibly unrelated to the outcome and is not expected to bias the estimates. Table 1 reports the affected sample counts.

[2]The remaining gaps are minor and isolated: a single day in June 2019, a cluster of days in January 2020, and scattered individual dates. Together they account for under two percent of days in the study period.

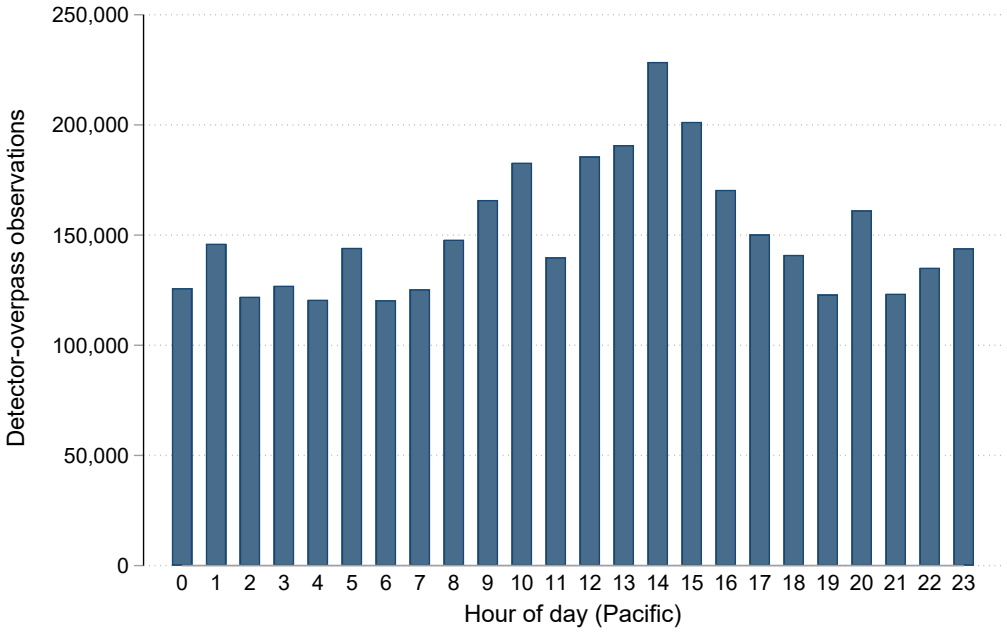
3.2 Land Surface Temperature: ECOSTRESS

Land surface temperature data come from the NASA ECOSTRESS `ECO_L2T_LSTE` Version 2 product, distributed by the NASA Land Processes Distributed Active Archive Center and accessed via AppEEARS (Hulley and Hook, 2022). ECOSTRESS is a thermal infrared sensor on the International Space Station (ISS) that has provided observations since July 2018 at 70-meter spatial resolution (Fisher et al., 2020). MODIS, common in urban heat research, observes at 1 kilometer (Chakraborty et al., 2019). ECOSTRESS's 70-meter resolution allows LST measurement at the scale of individual freeway segments and adjacent land parcels.

The defining temporal property of ECOSTRESS for this study is its ISS orbit. Sun-synchronous satellites overpass at a fixed local solar time (Landsat 8 at approximately 10am, MODIS at fixed morning and afternoon times). The ISS, by contrast, precesses through a range of local overpass times at any given California location. Successive ECOSTRESS overpasses of the same segment therefore occur at different times of day, generating exogenous variation in the hour of LST observation. Figure 2 plots the distribution of overpass times across the 24-hour day. We exploit this variation to match each LST observation to a contemporaneous PeMS congestion reading at the overpass hour. The hour-of-day variation in overpass time also underlies the event study evidence in Section 5.1.

We use three layers from ECO_L2T_LSTE: LST (in Kelvin), QC (quality control flags), and cloud (cloud mask). We apply quality filters to exclude pixels where the LST retrieval is unreliable. Three sources of retrieval failure motivate these filters. First, clouds absorb and re-emit thermal radiation before it reaches the sensor, so a cloudy pixel records the temperature of the cloud rather than the ground surface. We exclude pixels flagged as cloudy by the cloud mask. Second, the retrieval algorithm corrects the raw thermal signal for atmospheric absorption and scattering. This correction can fail under unusual water vapor or aerosol conditions (including, potentially, haze from heavily congested corridors), producing biased LST values. We exclude pixels with QC flag values indicating failed or degraded retrievals. Third, the algorithm converts radiance to temperature using an assumed

Figure 2: Distribution of ECOSTRESS overpass times by hour of day (Pacific) in the estimation panel



surface emissivity, so some retrievals are implausible. We drop pixels whose retrieved LST falls outside a physically plausible range (255–345 K, about −18 to 72°C), which removes implausible retrievals that survive the cloud and quality-control screens. Using corrupted pixels would add noise to the outcome variable and could introduce systematic bias if retrieval quality is correlated with congestion levels. After applying these filters, the outcome for observation (i, t) is the mean LST across all pixels passing quality filters within a distance band around detector i at overpass t (Section 3.3.1). The spatial footprint is restricted to a 2-kilometer buffer around the California state highway network, encompassing all PeMS detector locations and the immediately surrounding land.

Because ECOSTRESS is a relatively young sensor, coverage in the earliest months of our study period is incomplete. ECOSTRESS experienced two anomalies of its onboard mass storage unit that suspended data acquisition (NASA Land Processes Distributed Active Archive Center, 2024). The primary unit failed on September 29, 2018, and acquisition resumed only after the sensor was switched to its backup unit on December 5, 2018. A

second anomaly halted acquisition again between March 14 and May 15, 2019. Our panel contains no LST observations for October through December 2018 and reduced coverage in the spring of 2019. These gaps arise from sensor hardware status rather than from any feature of traffic, weather, or location. The lost observations are therefore unrelated to the outcome and affect only the early-period sample size, not the identifying variation. ECOSTRESS coverage is also sparse from January through March 2024, which leaves few observations in those months.

3.3 Sample Construction

The unit of observation is a freeway detector station i crossed with an ECOSTRESS overpass t. A valid observation requires at least five ECOSTRESS pixels passing quality filters within 500 meters of the detector, and a valid PeMS congestion reading within the same calendar hour as the overpass timestamp. That congestion reading must also pass a data-quality filter, with at least 50 percent of the hour's five-minute intervals observed, to reduce measurement error from detector malfunctions.

Table 1 reports the observations lost at each of these filtering stages. The binding step is the same-hour match, which retains about 46 percent of quality-passing overpasses. A freeway detector reports a usable speed only intermittently: some never report speed at all, and a working detector is online at a given overpass hour only about half the time. The matched panel is therefore the intersection of two independent data streams rather than a curated subsample. The match rate is essentially flat across all twenty-four hours of the day, so this attrition costs statistical power without selecting on the overpass-time variation the design exploits. Because detector reporting reliability and cloud cover are plausibly unrelated to the idiosyncratic timing of freeway accidents, we do not expect the attrition to threaten identification.

Table 2 reports summary statistics for the estimation panel and, separately, for the free-flow-onset-treated overpasses that identify the estimates. Treatment is rare by design:

accidents are infrequent, so only 5,096 of the 3.6 million detector-overpasses, at 2,816 distinct detectors, carry a free-flow-onset accident in the prior two hours. Overpasses with no accident in the prior two hours are the control pool: identification comes from comparing treated and control overpasses conditional on detector, date, and hour-of-day fixed effects. Two patterns are worth noting. The typical overpass is close to free-flow (mean speed performance index 0.92). The treated overpasses are modestly more congested than the estimation panel as a whole (index 0.90 versus 0.92, with higher traffic flow), the descriptive counterpart of the first stage. Surface temperature averages 22.7°C with wide dispersion (standard deviation 11.3°C), reflecting the all-hours, multi-year, statewide ECOSTRESS coverage that the detector, date, and hour-of-day fixed effects absorb.

Table 1: Sample construction

	Observations	Detectors	% of row 1
ECOSTRESS overpass observations	9,045,115	11,180	—
with ≥ 5 quality-passing pixels (500 m buffer)	8,910,223	11,178	98.5
one observation per detector and overpass	8,412,425	11,178	93.0
matched to a same-hour PeMS congestion reading	3,840,309	9,323	42.5
excluding low-coverage incident days	3,623,637	9,287	40.1
Estimation panel	**3,623,637**	**9,287**	40.1
with non-missing weather controls	3,466,297	8,889	38.3

Notes: The unit of observation is a detector × ECOSTRESS overpass. Row 1 counts overpasses with a valid LST extraction (July 2018–March 2026; after pixel-level cloud, quality-control, and 255–345 K masking). Row 3 keeps one observation where ECOSTRESS records a detector twice in one overpass. The two map tiles of one scene are averaged by pixel count, and of two overlapping scenes the one with more valid pixels is kept. Row 5 drops overpasses on dates with missing or anomalously sparse CHP incident records (the March–July 2025 feed gap and scattered days), which cannot be coded incident = 0. The final row is the panel with weather controls available.

3.3.1 Outcome: Distance Bands around the Detector

The outcome is land surface temperature in concentric distance bands around the detector, each the mean of quality-filtered ECOSTRESS pixels within a given radius at overpass t. We construct the 100-, 250-, 500-, and 1,000-meter buffers, and from them three disjoint annular rings (0–100, 100–500, and 500–1,000 meters) for the distance gradient. We call

Table 2: Summary statistics

	Full sample		Free-flow-onset treated	
	Mean	SD	Mean	SD
Land surface temp (°C)	22.70	11.29	22.33	10.84
Speed performance index (speed / free-flow)	0.92	0.13	0.90	0.15
Speed (mph)	63.50	8.81	62.37	10.67
Free-flow speed (mph)	68.77	1.96	69.01	1.48
Traffic flow (veh/hr)	2,741	2,070	3,054	1,935
ECOSTRESS pixels in buffer	143.3	34.7	143.5	34.5
Observations	3,623,637		5,096	

Notes: Unit of observation is a PeMS detector × ECOSTRESS overpass, July 2018–March 2026. "Full sample" is the estimation panel; "free-flow-onset treated" is the subsample of overpasses with a free-flow-onset accident (one striking the segment while it was free-flowing, pre-accident speed performance index ≥ 0.90) in the prior two hours. The speed performance index is realized speed divided by free-flow speed (1 = free-flow; lower = more congested). Land surface temperature and pixel counts refer to the 500-meter buffer.

the innermost ring the near-road ring and the 500–1,000-meter ring the outer ring. The band spacing follows the near-road pollution literature, which identifies an exposure zone extending up to 300 to 500 meters from a major road (Health Effects Institute, 2010). We use those distances to set the geometry of the bands, not as evidence about how far heat itself travels. A pollutant disperses as a near-conserved tracer, while heat is shed to the air and to the sky far faster. The freeway runs through every band, but the outer bands mostly capture temperature in the surrounding area, where people live. At 70-meter resolution a 500-meter radius contains about 160 pixels, yielding a stable average; we take the 500-meter buffer as the descriptive baseline, while the preferred causal estimate uses the near-road ring. Section 5.2 tests how the effect changes across the rings.

4 Research Design

Our design uses freeway accidents as an instrument for congestion. Because congestion itself raises accident risk, the design is valid only if accident timing is exogenous to congestion. An event study tests this first. The reduced form, our primary estimate, then measures the effect

of an accident on surface temperature, and 2SLS scales that effect per unit of congestion.

4.1 Event Study

The event study traces congestion and surface temperature in two-hour bins around an accident. It asks whether congestion is already building before impact (reverse causality), and whether pre-accident trends in surface temperature would violate the exclusion restriction.

For each ECOSTRESS overpass we measure the elapsed time to each qualifying accident on the same detector within a ±6-hour window and assign the overpass to a two-hour lag bin $[\ell, u)$:

$$d_{\ell,u,it} = \mathbf{1}[t - \text{accident}_{it} \in [\ell, u)] \,.$$

Negative lags are overpasses before an accident; the baseline group has no accident on the segment within ±6 hours. We estimate

$$Y_{it} = \alpha_i + \mu_d + \lambda_h + \sum_{[\ell,u)} \phi_{\ell,u}\, d_{\ell,u,it} + \mathbf{X}'_{it}\gamma + \eta_{it} \tag{1}$$

separately for $Y_{it} = \text{SPI}_{it}$, the speed performance index on the segment, and $Y_{it} = \text{LST}_{it}$, surface temperature in degrees Celsius in the near-road ring (0–100 meters from the detector; Section 3.3.1). Here α_i are detector fixed effects, μ_d are date fixed effects (where d indexes the calendar date of overpass t), and λ_h are hour-of-day fixed effects (where h indexes the Pacific-time hour of the overpass, $h \in \{0, 1, \ldots, 23\}$). The vector $\mathbf{X}_{it}$ holds weather controls, and standard errors are clustered at the detector. The lead coefficients $\phi_{\ell,u}$ at negative lags are the diagnostic of interest, in speed-performance-index units for congestion (the index averages about 0.92) and in degrees Celsius for temperature.

Identification is within-segment, within-date, and within-hour. The detector effects α_i absorb time-invariant location characteristics (pavement, land cover, urban density). The date effects μ_d absorb common daily shocks such as heat waves without eliminating within-day variation in overpass time, and the hour effects λ_h absorb the solar cycle. The controls

$\mathbf{X}_{it}$ are precipitation, relative humidity, and wind speed at the detector and overpass hour, from the ERA5-Land hourly reanalysis (Muñoz Sabater et al., 2021; Muñoz Sabater, 2019). They absorb weather not captured by the date fixed effects. ERA5-Land's 9-kilometer grid has no urban land cover and is too coarse to register an accident's congestion. We omit air temperature because it closely tracks the outcome; Table 3 also reports the estimates with it included.

To separate exogenous from congestion-induced accidents, we classify each accident by the speed performance index on its segment in the pre-accident hour. A free-flow-onset accident strikes while the segment is free-flowing (pre-accident SPI ≥ 0.90), so no congestion on the segment precedes it; a congested-onset accident strikes while the segment is already congested. Among accidents we can classify, the two groups are roughly equal in number. We estimate equation (1) with separate bin sets for the two groups in a single regression, so that congested-onset overpasses do not enter the free-flow-onset baseline. Section 5.1 reports the results.

4.2 Reduced Form and Two-Stage Least Squares

Even conditional on these fixed effects and controls, SPI_{it} is endogenous. Clear skies raise both solar load and traffic volumes, biasing OLS toward a warming effect, while heat-induced travel avoidance works in the opposite direction, so the sign of the net bias is ambiguous ex ante.

Our instrument for SPI_{it} is an indicator for a free-flow-onset accident on segment i in the hours before the overpass, built in two steps. First, we flag any qualifying accident (Section 3.1) on segment i within a lookback window of length H, with $H = 2$ hours in the baseline. Second, we keep only accidents that struck while the segment was free-flowing, defined as a speed performance index of at least 0.90 in the pre-accident hour, so that no

congestion on the segment precedes the accident:

$$Z_{it} = \mathbf{1}[\text{free-flow-onset accident on segment } i \text{ in } [t-H, t]]\,.$$

Restricting to free-flow-onset accidents removes the reverse causality the event study reveals (Section 5.1). The two-hour baseline matches the window over which the event study concentrates the congestion shock and its temperature response. We report $H \in \{1, 3\}$ hours as robustness (Table A.9, panel B) and vary the 0.90 onset threshold in Section 6.

The instrument must also be relevant, predicting congestion on the affected segment. Accidents reduce effective capacity, slowing traffic as vehicles merge around the incident. The resulting slowdown can last almost three hours and spread several kilometers (Cao et al., 2021), and its duration need not match the logged clearance time (Haule et al., 2019). Free-flow-onset accidents perturb congestion less than the average accident (they strike uncongested segments), so their first stage is smaller in magnitude, though still precisely estimated. We report the Kleibergen–Paap first-stage F (KP F in the tables) with each specification. A severity check (Table A.3) weights the instrument by accident duration, since longer accidents sustain larger congestion.

Our primary estimate is the reduced form, which regresses surface temperature directly on the instrument:

$$\text{LST}_{it} = \alpha_i + \mu_d + \lambda_h + \pi\, Z_{it} + \mathbf{X}'_{it}\gamma + e_{it} \tag{2}$$

The coefficient π is the effect of a free-flow-onset accident on surface temperature in degrees Celsius. To express it per unit of congestion, the first stage regresses the speed performance index on the instrument and the fixed effects and controls, and the second stage replaces it

with its fitted value:

$$\text{SPI}_{it} = \alpha_i + \mu_d + \lambda_h + \beta\, Z_{it} + \mathbf{X}'_{it}\gamma + \varepsilon_{it} \tag{3}$$

$$\text{LST}_{it} = \alpha_i + \mu_d + \lambda_h + \delta\, \widehat{\text{SPI}}_{it} + \mathbf{X}'_{it}\gamma + u_{it} \tag{4}$$

We expect $\beta < 0$: accidents raise congestion, lowering the index. The coefficient δ is the effect of a one-unit increase in the speed performance index on LST in degrees Celsius. Because congestion lowers the index, warming from congestion appears as $\delta < 0$, and $-\delta$ is the effect of a shift from free-flow to fully congested conditions. We also report the implied effect of a 10 mph speed reduction (the unit used by Lee and Berkelhammer, 2025). Because a free-flow-onset accident lowers the index by only about 0.0154 (roughly 1 mph), the full one-unit quantity is an extrapolation far beyond the identified margin. We estimate the reduced form and the second stage for each distance band: Section 5.2 traces the reduced form from the roadway out to the surrounding area, and Section 5.3 reports the 2SLS estimates.

The exclusion restriction requires that accident occurrence affects LST only through its effect on traffic congestion, conditional on the fixed effects and controls. We identify three potential violations and address each empirically; the event study (Section 5.1) provides the primary test, with the robustness checks in Section 6 providing supporting evidence.

Volume-driven clustering. Accident probability rises with traffic volume, which is correlated with congestion. The fixed effects absorb the systematic component of this relationship, leaving the question of whether within-segment, within-date accident timing correlates with temperature determinants beyond what they and the weather controls capture. The event study's flat pre-accident LST coefficients are the central piece of supporting evidence (there is no anticipatory movement in the outcome). We omit contemporaneous traffic volume because it is jointly determined with congestion and itself reduced by accidents, a bad control that would absorb part of the causal pathway (Angrist and Pischke, 2009).

Weather. Precipitation raises accident rates and lowers LST through evaporative

cooling and reduced solar radiation. Conditional on the fixed effects, free-flow-onset accidents are slightly more frequent in warmer and more humid hours (Table A.8). The specification controls for humidity, wind speed, and precipitation; panel B of Table 3 also controls for air temperature.

Direct thermal effects and incident classification. Vehicle fires emit heat directly at the road surface, elevating pixel temperatures through a non-congestion channel. We exclude vehicle fires. For qualifying incidents (collisions, hit-and-run collisions, and spinouts), the crash scene, including any emergency response, is a point source. It lies within 100 meters of the matched detector for about a quarter of free-flow-onset accidents, half of which last 30 minutes or less. An active scene therefore sits inside the near-road ring at the time of the overpass in only a minority of treated observations. Stopped traffic in the queue is a separate channel, which we bound in Section 6.

We estimate the model with `ivreghdfe` (Correia, 2018), which extends `ivreg2` (Baum et al., 2007), and cluster standard errors at the detector. The reduced form's test of a zero effect does not depend on the strength of the first stage. It is the Anderson–Rubin test (Andrews et al., 2019), so the reduced form and the 2SLS estimate share one test of a zero effect. We also report the Anderson–Rubin confidence set for the 2SLS estimate (Section 5.3).[3]

5 Results

The estimation panel holds roughly 3.6 million detector-overpass observations (Section 3). We report three results: the near-road temperature response to a free-flow-onset accident, how that response changes with distance from the road, and what it implies per unit of congestion.

[3]Because two-stage least squares with non-saturated covariates need not recover a positively weighted average of treatment effects (Blandhol et al., 2022), we treat δ as a local, design-based summary rather than a structural parameter.

5.1 Near-Road Temperature Response

The quantity we emphasize is the reduced form: a free-flow-onset accident raises near-road surface temperature by about 0.11°C over the following two hours (0.114, SE 0.047, $p = 0.01$), against flat pre-accident leads. With air temperature added to the controls the estimate is 0.08°C (0.081, $p = 0.06$; panel B), and without weather controls it is 0.09°C (Table A.5). For scale, on overpasses without an accident the near-road ring averages 0.21°C warmer than the outer ring, so the accident adds about half of that difference. The reduced form is the direct contrast (treated versus untreated overpasses, conditional on detector, date, and hour-of-day fixed effects). Unlike the per-unit estimate, it does not depend on how the first stage is measured. Table 3 reports it by distance ring.

Table 3: Reduced form: effect of a free-flow-onset accident on land surface temperature (°C), by distance ring

	(1) 0–100 m (near-road)	(2) 100–500 m	(3) 500–1000 m
A. Weather controls: relative humidity, wind speed, precipitation			
Free-flow-onset accident (prior 2h)	0.1143**	0.0931**	0.0561
	(0.0467)	(0.0453)	(0.0466)
B. Adding air temperature			
Free-flow-onset accident (prior 2h)	0.0808*	0.0601	0.0201
	(0.0435)	(0.0419)	(0.0434)
Free-flow-onset treated	4,642	4,600	4,807
Observations	3,304,407	3,264,271	3,419,166

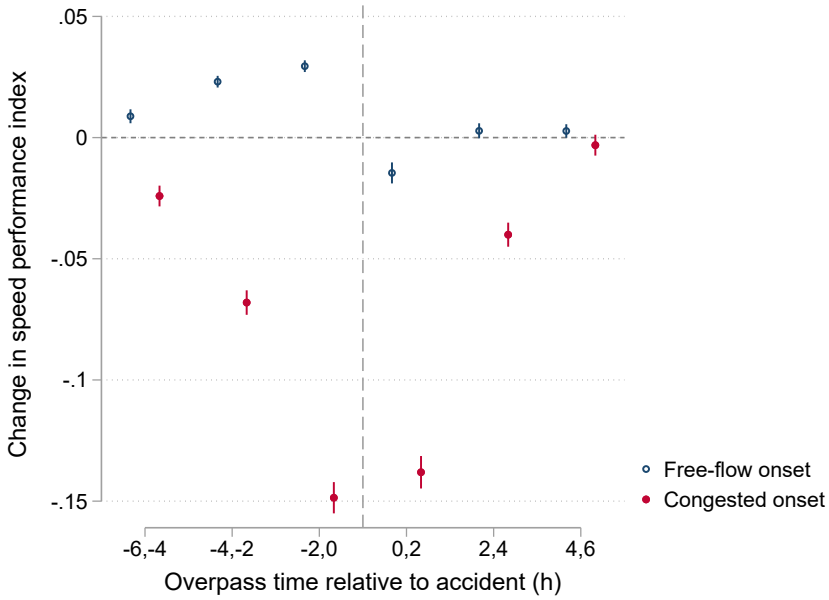
Notes: Reduced-form regressions of ring-mean LST (°C) on the free-flow-onset accident indicator, on the sample pairing free-flow-onset-treated overpasses with no-accident controls. Panel A controls for relative humidity, wind speed, and precipitation at the detector and overpass hour; panel B adds air temperature. Both panels use the same observations. All columns include detector, date, and hour-of-day fixed effects. Observations differ across columns because each ring requires its own valid pixels. Specifications without weather controls are in Table A.5. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The event study traces this response over time (Figure 3). The two onset groups behave very differently in congestion (Figure 3a). Before a congested-onset accident, congestion ramps steeply (−0.024, −0.068, −0.149 across the three lead bins): the reverse causality is concentrated here. Before a free-flow-onset accident, congestion does the op-

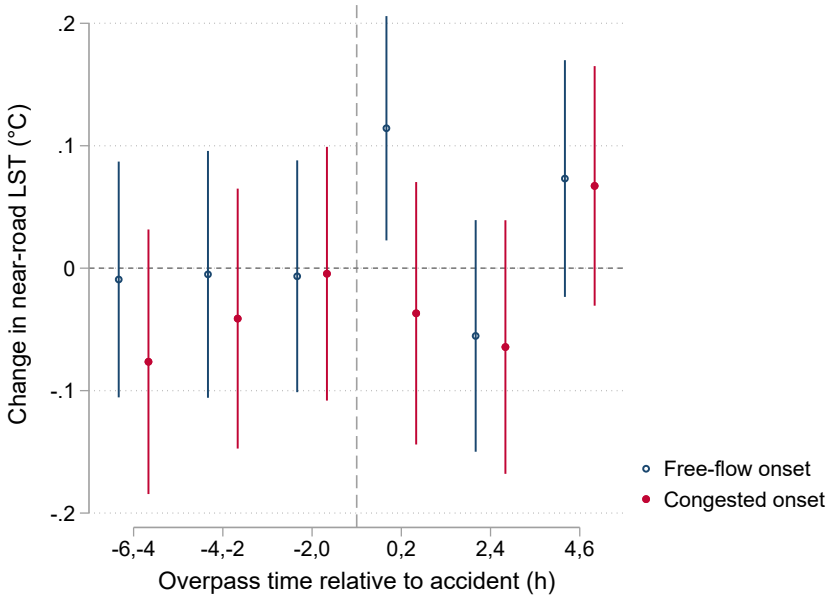
posite: the segment is, if anything, increasingly free-flowing (+0.009, +0.023, +0.029) and then falls at impact (−0.015, $t \approx -7$). Free-flow-onset accidents thus deliver a small, discontinuous congestion shock with no congestion building beforehand. The upward drift in those leads reflects the persistence of the free-flowing state on which the instrument conditions, since the pre-accident hour itself falls in the final lead bin. Section 5.3 measures this persistence directly with a pseudo-event placebo that reproduces the conditioning without an accident. What the leads establish is that congestion is not building before these accidents, the reverse causality the restriction is designed to remove. Surface-temperature leads for free-flow-onset accidents are individually insignificant and jointly indistinguishable from zero ($F(3,\ 8{,}849) = 0.02$, $p > 0.99$), consistent with the exclusion restriction. For congested-onset accidents, which the instrument excludes, they are jointly insignificant ($p = 0.50$; Figure 3b). The free-flow-onset impact bin matches the reduced form (+0.114°C). By two to four hours after the accident the speed performance index is back at its baseline, so the later temperature bins carry no congestion contrast. They alternate in sign, and neither is significant (−0.055 and +0.073). The onset event study in the 500-meter buffer without weather controls also shows flat free-flow-onset temperature leads ($F(3,\ 9{,}248) = 0.22$, $p = 0.88$; Figures A.3 and A.4).

For reference, an event study on all accidents shows congestion building before impact, the reverse causality that the free-flow-onset restriction removes, while surface-temperature leads stay flat (Figures A.1 and A.2). In the 500-meter buffer without weather controls (Table A.2), the OLS association between the speed performance index and surface temperature is small and wrong-signed (+0.10). The all-accident 2SLS estimate, contaminated by that reverse causality, is −1.17 (−1.22 with traffic volume added), against −4.63 for free-flow-onset accidents in the same specification.

Figure 3: Event study by accident onset



(a) Speed performance index



(b) Near-road surface temperature (°C)

Notes: Coefficients $\hat{\phi}_{\ell,u}$ from equation (1) for free-flow-onset accidents (segment free-flowing in the pre-accident hour) and congested-onset accidents, estimated jointly: (a) the speed performance index; (b) surface temperature in the near-road ring (0–100 meters). Detector, date, and hour-of-day fixed effects and weather controls; 95% confidence intervals clustered at the detector; baseline = no accident within ±6 hours.

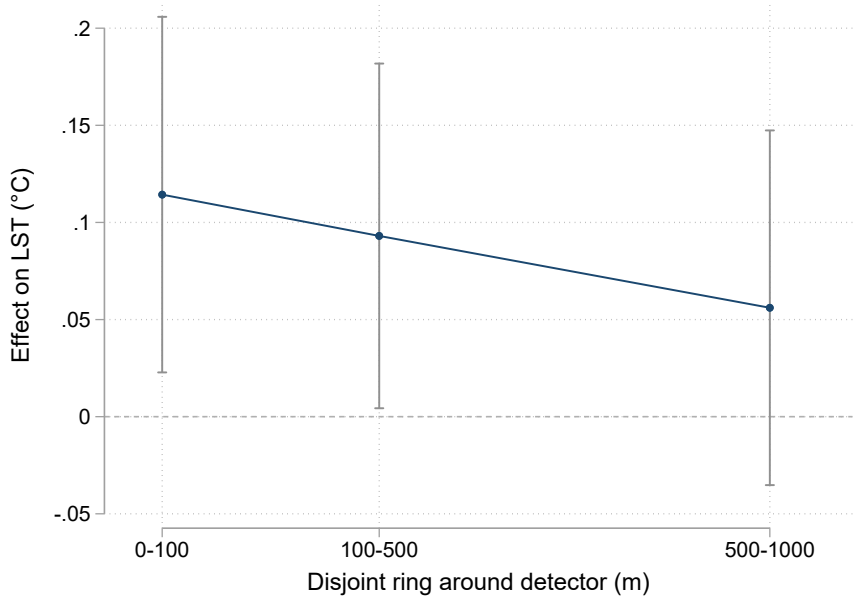
5.2 Distance Decay

Table 3 and Figure 4 report the reduced form by distance from the detector. This distance gradient is the paper's spatial object, the thermal analog of the distance decay traced for transport and environmental disamenities (Brinkman and Lin, 2024; Currie et al., 2015). Because the rings are disjoint, each estimate reflects the temperature response in a distinct distance zone, and the gradient is not mechanically smoothed by sharing near-road pixels across bands. The near-road ring captures on-road and roadside temperature, and the outer ring mostly captures the area around the corridor. The gradient adjudicates whether a measurable effect persists out to the outer ring, and whether it decays with distance or instead rises in the intermediate zone. A rise there would point to traffic diverted onto surface streets.

The response decays monotonically with distance from the detector. A free-flow-onset accident raises surface temperature by 0.114°C in the near-road ring ($p = 0.01$), 0.093°C at 100–500 meters ($p = 0.04$), and 0.056°C in the 500–1,000-meter outer ring ($p = 0.23$). The effect is right-signed in every ring and significant at the 5 percent level in the two inner rings. The outer-ring estimate is of the expected sign, but its confidence interval contains both zero and an effect as large as the near-road estimate. The response is concentrated at the road and falls off with distance, with no rise in the intermediate zone of the kind a surface-street diversion effect would produce. Whether the decay itself is statistically real, rather than a pattern read off separate confidence intervals, is tested formally below.

Table 4 carries out that test. It estimates the three ring reduced forms jointly, stacking the disjoint rings and fully interacting the specification (fixed effects, weather controls, and the accident indicator) with ring. Standard errors are clustered on a 5-kilometer spatial grid, so nearby detectors, and the three rings of the same detector-overpass, may be arbitrarily correlated. The test is cast in reduced form, consistent with the paper's emphasis: the ring coefficients are the estimates in panel A of Table 3, in °C per accident window. Although the per-ring confidence intervals overlap, the cross-ring differences are estimated much more

Figure 4: Effect of a free-flow-onset accident on surface temperature by distance from the detector



Notes: Reduced-form estimates (°C) with weather controls for disjoint annular rings 0–100, 100–500, and 500–1,000 meters from the detector, with 95% confidence intervals (standard errors clustered by detector). Point estimates match panel A of Table 3.

precisely than the ring levels, because the rings of a given detector-overpass share common shocks that difference out of the comparison. The data reject equality of the three ring effects ($p = 0.024$) and reject equality of the near-road and outer rings ($p = 0.008$); a linear decline in the effect with distance from the detector is likewise significant ($p = 0.009$). The distance decay is thus a statistically established feature of the reduced form, not a pattern inferred from overlapping point estimates. Because a broad area-level confounder would move all rings together rather than fade with distance from the detector, the rejected equality shows that the decline is not an artifact of such confounding. The test sharpens, rather than changes, the substantive reading: the externality demonstrably weakens with distance from its source, while the level of the response within the outer ring itself remains imprecisely estimated.

How many people live in these rings? Table 5 reports the residential population around the studied freeway segments from the 2018–2022 American Community Survey.

Table 4: Tests of the reduced-form distance gradient

Free-flow-onset accident × 0–100 m (near-road)	0.1143**
	(0.0474)
Free-flow-onset accident × 100–500 m	0.0931**
	(0.0458)
Free-flow-onset accident × 500–1000 m (outer)	0.0561
	(0.0481)
p: all three rings equal	0.024
p: near-road = outer	0.008
Distance trend (°C per 100 m)	−0.0083
p: distance trend	0.009
Observations (stacked)	9,987,844

Notes: The three disjoint rings are stacked and estimated jointly, with the fixed effects (detector, date, hour-of-day), the weather controls, and the accident indicator each fully interacted with ring; the ring coefficients therefore reproduce the reduced forms in panel A of Table 3. Standard errors, in parentheses, are clustered on a 5-kilometer spatial grid (945 clusters), allowing arbitrary correlation across nearby detectors and across the rings of the same detector-overpass. Equality rows report Wald tests on the joint covariance matrix; the distance-trend rows replace the ring-specific coefficients with the accident indicator interacted with ring-midpoint distance. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

About 11 million Californians live within one kilometer of these detectors, and 4.7 million within 500 meters. This population is more racially diverse than the state as a whole (42 percent Hispanic and 70 percent non-white, against roughly 40 and 65 percent statewide). Its median household income is close to the statewide figure. This describes the scale and composition of the exposed population, not how the effect is distributed across it.

5.3 Per Unit of Congestion

Table 6 expresses the response per unit of congestion by dividing each ring's reduced form by its first stage. Near the road, a free-flow-onset accident lowers the speed performance index by 0.0154 ($F \approx 48$), less than the average accident does (Table A.1, without weather controls). The resulting 2SLS estimate is $\delta = -7.41$ (SE 3.20, $p = 0.02$). The Anderson–Rubin 95% set, which is valid regardless of instrument strength, is $[-14.6, -1.5]$; its test of a zero effect is the reduced-form test ($p = 0.01$). The estimate is right-signed across alternative

Table 5: Residential population exposed near the studied freeway segments

Distance ring	Exposed pop.	Median income	% Hispanic	% non-white
0–100 m	319,927	$100,193	42	71
100–500 m	4,413,002	$100,329	42	71
500–1000 m	6,273,904	$102,833	41	70
Corridor (0–1000 m)	11,006,834	$101,752	42	70
California	39,356,104	$104,733	40	65

Notes: Population, median household income, and racial composition from the 2018–2022 ACS five-year block-group estimates, area-weighted into disjoint rings by distance to the nearest detector in the estimation panel. Detector buffers are unioned across segments, so each resident is counted once. "% non-white" is one minus the non-Hispanic white share; "Median income" is the population-weighted mean of block-group median household incomes. The California row is the statewide population-weighted benchmark.

instruments and outcome bands, though significance weakens in the wider bands (Tables A.3 and A.4). We read this per-unit coefficient as a calibration of the reduced form.

We measure the persistence of the selected free-flowing state with a placebo that replicates the instrument's construction without an accident. For each real free-flow-onset accident we assign a pseudo-event at the same timestamp to a randomly drawn other detector that was free-flowing in the pre-accident hour (SPI ≥ 0.90) and had no accident within six hours. The 283,236 pseudo-events reproduce the instrument's date and time-of-day distribution almost exactly. They share each accident's date and time but not its detector. A match on the same detector and pre-accident state would instead compare overpasses on different dates or at different hours, adding day-to-day and diurnal temperature swings that the date and hour-of-day fixed effects absorb. Table 7 reports their effects on the no-accident sample. The pseudo first stage is positive ($+0.0207$, $t = 20$): a segment selected on a free-flowing pre-accident hour remains more free-flowing than its baseline over the following two hours, rather than reverting below it. The accident-induced drop to 0.0154 below baseline therefore cannot be mean reversion, and measured against this no-accident counterfactual, the accident lowers the index by about 0.036 rather than 0.0154. The pseudo reduced forms on temperature are null at both bands: -0.037 in the near-road ring ($p = 0.45$) and -0.001 in the 500-meter buffer ($p = 0.99$). Conditioning on a free-flowing pre-accident hour has no

Table 6: Reduced form, first stage, and 2SLS by distance ring

	(1) 0–100 m	(2) 100–500 m	(3) 500–1000 m
A. Weather controls: relative humidity, wind speed, precipitation			
Reduced form: accident → LST (°C)	0.1143**	0.0931**	0.0561
	(0.0467)	(0.0453)	(0.0466)
First stage: accident → SPI	-0.0154***	-0.0153***	-0.0147***
	(0.0022)	(0.0022)	(0.0022)
2SLS: effect of SPI on LST (°C)	-7.409**	-6.077**	-3.818
	(3.195)	(3.085)	(3.217)
Anderson–Rubin 95% set (2SLS)	[-14.6, -1.5]	[-13.0, -0.3]	[-10.8, 2.4]
KP first-stage F	47.8	46.6	45.4
B. Adding air temperature			
Reduced form: accident → LST (°C)	0.0808*	0.0601	0.0201
	(0.0435)	(0.0419)	(0.0434)
First stage: accident → SPI	-0.0154***	-0.0153***	-0.0147***
	(0.0022)	(0.0022)	(0.0022)
2SLS: effect of SPI on LST (°C)	-5.229*	-3.921	-1.369
	(2.896)	(2.788)	(2.957)
Anderson–Rubin 95% set (2SLS)	[-11.6, 0.3]	[-10.0, 1.5]	[-7.5, 4.6]
KP first-stage F	47.9	46.7	45.5
Free-flow-onset treated	4,642	4,600	4,807
Observations	3,304,407	3,264,271	3,419,166

Notes: Each column is a disjoint ring around the detector. The reduced form regresses ring-mean LST on the free-flow-onset accident indicator; the first stage regresses the speed performance index on the same indicator; the 2SLS coefficient is their ratio, the effect of a one-unit increase in the speed performance index on LST (warming from congestion appears as a negative coefficient). Panel A controls for relative humidity, wind speed, and precipitation; panel B adds air temperature. All rows include detector, date, and hour-of-day fixed effects, on the sample pairing free-flow-onset-treated overpasses with no-accident controls. The first stage differs slightly across columns because the ring samples differ. Standard errors, clustered by detector, in parentheses. The Anderson–Rubin set is the 95% confidence set for the 2SLS coefficient that remains valid regardless of instrument strength. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

detectable effect on surface temperature, consistent with the exclusion restriction: the temperature response follows the accident rather than the selection. The placebo also quantifies the first-stage accounting that motivates reporting the per-unit magnitude as a range.

Table 7: Pseudo-event placebo: speed performance index and land surface temperature

	(1) Speed performance index	(2) LST, near-road ring (°C)	(3) LST 500 m (°C)
Pseudo free-flow-onset event	0.0207***	-0.0369	-0.0006
	(0.0011)	(0.0490)	(0.0489)
Pseudo-event treated	5,287	5,046	5,287
Observations	3,456,633	3,299,765	3,456,633

Notes: Each real free-flow-onset accident is assigned a pseudo-event at the same timestamp on a randomly drawn other detector that was free-flowing in the pre-accident hour and had no accident within six hours, replicating the instrument's construction without an accident. Columns report regressions of the stated outcome on the pseudo-event indicator, estimated on the no-accident sample, with weather controls. Detector, date, and hour-of-day fixed effects; standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The per-unit scaling depends on how much of the instrument's association with congestion is credited to the accident shock rather than to the selection inherent in conditioning on a free-flowing pre-accident hour. That choice must be applied to both sides of the ratio. Taking the measured quantities at face value, the 0.1143°C reduced form over a first stage of -0.0154 gives $\delta \approx -7.41$, the 2SLS estimate above. Substituting the selection-adjusted first stage from the pseudo-event placebo (Table 7; $-0.0154 - 0.0207 = -0.036$) requires also netting out the placebo's own reduced form (-0.037 in the near-road ring). Netting it out raises the numerator to 0.151°C and gives $\delta \approx -4.2$. At a mean free-flow speed of 68.8 mph, 10 mph is 0.145 index units, so the two accountings imply roughly 0.6 to 1.1°C per 10 mph of slowdown. With air temperature also controlled, the two accountings give 0.4 to 0.8°C. Both endpoints exceed the 0.36°C that Lee and Berkelhammer (2025) report for Chicago bus speeds, but the two are not like-for-like: their outcome is a two-kilometer heat-island average, and their design is associational. Because the placebo reduced form is imprecisely estimated ($p = 0.45$), we report the two accountings as a range. For reference, the average weekday slowdown at the studied detectors is 9.7 mph in the morning and

evening peak hours. The estimate is also local: it describes segments that were free-flowing in the pre-accident hour and then slowed suddenly. Chronically congested segments enter the sample whenever they are free-flowing before an accident. An accident can only slow traffic where there is room to slow it, however, so the first stage comes from segments with slack capacity.[4] The estimate therefore speaks to the marginal heat effect of an unusual slowdown rather than to the steady-state heat of a chronically congested freeway, which a within-detector design does not identify.

6 Robustness Checks

This section reports the checks most central to the instrument's construction and to what the near-road estimate measures. These are the sensitivity of the estimate to the free-flow-onset threshold and a bound on how much of the near-road response could be vehicles in the pixel. Further checks are summarized at the end of the section and reported in the Appendix. Unless noted otherwise, each check re-estimates the preferred specification (near-road ring, weather controls, and detector, date, and hour-of-day fixed effects). The estimate is right-signed throughout.

Free-flow-onset threshold sensitivity. The instrument conditions on a lagged value of the endogenous variable: an accident qualifies only if its segment was free-flowing (SPI ≥ 0.90) in the pre-accident hour. Because this selects on transitory traffic states, the choice of threshold matters in two directions. Loosening it readmits accidents that congestion itself may have caused, reintroducing the reverse causality the restriction exists to remove. Tightening it conditions on increasingly unusual hours, raising the worry that the selected moments are transitorily atypical in temperature as well, so that mean reversion in the outcome masquerades as a treatment response. Table 8 re-estimates the preferred specification at

[4] Splitting detectors into quintiles by the share of their overpasses with an index below 0.90, the first stage of the preferred specification is between -0.012 and -0.029 in the four less congested quintiles and $+0.012$ (SE 0.006) in the most congested, where conditioning on a free-flowing pre-accident hour selects hours that are unusually free-flowing for that segment.

onset thresholds of 0.85, 0.90 (the baseline), and 0.95. The reduced form is positive at every threshold: 0.097°C at 0.85 ($p = 0.02$), 0.114 at the baseline, and 0.131 at 0.95 ($p = 0.02$). The first stage shrinks from -0.024 to -0.008 as the treated sample narrows to the most strictly free-flowing segments, and the first-stage F falls from 137 to 48 to 10. The 2SLS estimate, the ratio of the two, is therefore right-signed but sensitive: -4.1 at 0.85, -7.41 at the baseline, and -15.9 ($p = 0.06$) at 0.95. Against the mean-reversion reading, the event-study temperature leads for free-flow-onset accidents are flat individually and jointly ($p > 0.99$; Section 5.1). At the baseline threshold, treated overpasses are therefore not selected on transitorily warm or cool moments, so there is no temperature reversion for the post-accident bin to inherit.

The sign is robust to the onset definition; the per-unit magnitude is less so, which is one reason the paper leads with the reduced form.

Table 8: Reduced form, first stage, and 2SLS by free-flow-onset threshold

	(1) SPI $\geq$ 0.85	(2) SPI $\geq$ 0.90	(3) SPI $\geq$ 0.95
Reduced form: accident $\rightarrow$ LST (°C)	0.0973**	0.1143**	0.1310**
	(0.0423)	(0.0467)	(0.0557)
First stage: accident $\rightarrow$ SPI	-0.0236***	-0.0154***	-0.0082***
	(0.0020)	(0.0022)	(0.0026)
2SLS: effect of SPI on LST (°C)	-4.123**	-7.409**	-15.921*
	(1.823)	(3.195)	(8.461)
KP first-stage F	136.7	47.8	9.9
Free-flow-onset treated	5,800	4,642	2,952
Observations	3,305,565	3,304,407	3,302,717

Notes: The preferred specification (binary free-flow-onset instrument, LST in the near-road ring, weather controls) re-estimated with the onset threshold, the pre-accident speed performance index above which an accident qualifies as free-flow-onset, set to each stated value (0.90 is the baseline). The reduced form regresses LST on the free-flow-onset indicator; the first stage regresses the speed performance index on the same indicator; the 2SLS coefficient is their ratio. Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Is the near-road effect just stopped vehicles in the pixel? Because ECOSTRESS records the radiometric temperature of whatever fills a pixel, a queue of sun-heated vehicles could

raise the measured near-road temperature without warming the surrounding environment. A back-of-envelope calculation (Table A.10) bounds this channel. Free-flow-onset accidents barely slow traffic, so they add only a few vehicles to the near-road ring. Even generously hot vehicle surfaces raise the pixel mean by only about 0.014°C under the measured first stage. Vehicle composition can thus account for about 12 percent of the near-road response under the measured accounting, and about 21 percent under the selection-adjusted one. Under that accounting, a larger first stage implies more marginal vehicles but also a larger reduced form. On either accounting the bulk of the response is genuine warming of the road environment (0.10°C of the measured 0.114, 0.12°C of the adjusted 0.151), consistent with the waste-heat mechanism.

Further checks. A prior-day placebo replaces the instrument with a free-flow-onset accident at the same time of day on the previous day. It gives a negative coefficient on surface temperature, significant at the 5 percent level (-0.099, $p = 0.04$; Table A.6); with air temperature also controlled it is -0.077 ($p = 0.08$). Its small positive coefficient on the speed performance index ($+0.0090$) is consistent with the selection on free-flowing conditions built into the instrument. Because adjacent detectors can share ECOSTRESS pixels, we also cluster at coarser spatial units. The standard error rises only modestly, from 3.20 at the detector to 3.37 at the freeway corridor (Table A.7), and to 3.35 with two-way clustering by detector and date. The estimate stays right-signed with the accident's duration as an alternative or additional instrument (Table A.3). The Hansen J test does not reject the overidentified version ($p = 0.74$), though it has little power. The estimate also stays right-signed, though about half as large (-3.53), with both the outcome and the speed performance index demeaned within three-kilometer cells. We demean both because 87 percent of a detector's cell-mates lie on the same freeway, at a median distance of 938 meters, and are themselves partly treated. Leaving each detector out of its own cell mean gives -3.00. Table A.9 collects six further specifications. Downwind and upwind half-disks of the 500-meter buffer give similar, right-signed estimates (panel A). The estimate is smaller but

right-signed with one- and three-hour lookback windows (panel B). It stays right-signed after dropping control segments within two detector stations of an accident, which diverted traffic could affect (panel C). The same holds with a stricter detector data-completeness threshold (panel D) and with the nearest pixel or the 500-meter-buffer maximum as the outcome (panels E and F). Table A.11 splits the sample by the air temperature at the overpass. The reduced form is similar in every quartile (0.06 to 0.10°C), with intervals too wide to rule out large differences, and the first stage shrinks from −0.024 in the coolest quartile to −0.010 in the hottest: accidents on hot overpasses produce less congestion.

7 Conclusion

We estimate the short-run effect of freeway congestion on nearby surface temperature in California. Freeway accidents serve as an instrument for congestion, and the outcome is ECOSTRESS land surface temperature, matched to congestion at the same hour. The central estimate is the reduced form: a free-flow-onset accident raises near-road surface temperature by about 0.11°C over the following two hours, against flat pre-trends. Expressed per unit of congestion, the response implies roughly 0.6 to 1.1°C per 10 mph of speed reduction. The effect decays across the distance rings (a decline that is statistically significant when the rings are estimated jointly), and it remains right-signed in the outer ring.

A lesson that travels beyond this application concerns identification. Accidents are treated as exogenous instruments, yet they are not exogenous to congestion. An event study shows that congestion rises in the hours before a typical accident, so an instrument built from all accidents embeds reverse causality and yields a precise but contaminated estimate. Wherever events are used as instruments for the very conditions that make those events more likely, the same bias arises. Conditioning on the pre-event state (here, restricting to accidents that strike a free-flowing road) is a transparent remedy, at the cost of a smaller first stage.

Because the estimates are identified from accident-induced episodes, they bear on what incident detection and clearance programs mitigate. Beyond delay and emissions, these episodes raise surface temperature, on average, on the road and its immediate surroundings, a thermal effect absent from standard accounts of congestion externalities.

The estimates have limitations. The per-unit magnitude is identified from a modest congestion shock, so it is not tightly pinned down. The near-road level also depends on the weather controls (0.08°C when air temperature is included), and the temperature response in the outer ring is not statistically resolved. Nor can we resolve whether the response is larger on hot days, where the mortality literature places the damage (Barreca et al., 2016; Heutel et al., 2021): the hottest cells hold few treated overpasses, and on hot afternoons accidents produce little congestion to begin with. Our evidence speaks to where the response is, not to what carries it. The gradient shows no rise of the kind surface-street diversion would produce, but we do not establish a physical channel by which roadway heat would reach the outer rings. The estimates are local to California freeway segments with PeMS detectors and to the variation induced by free-flow-onset accidents, which arises where traffic has room to slow. They need not extend to other settings or to longer-run or larger congestion changes, and they do not measure the steady-state heat of a chronically congested corridor.

Within those bounds, the paper contributes a causal estimate of the congestion-heat relationship, and a transparent way to handle reverse causality in accident-based instruments. It also contributes a spatial measurement. Resolving the response in distance bands shows how a roadway heat externality declines with distance (some 11 million Californians live within a kilometer of the studied detectors). The measurement speaks to residential proximity to freeways and the spatial distribution of urban heat exposure. Mortality during United States heat waves rises by about 2.5 percent for each additional degree Fahrenheit of the wave's mean temperature (Anderson and Bell, 2011), so on such days even a small increment in air temperature carries a cost; whether the surface response documented here carries into air temperature is not established. Whether the responses documented here are

large enough to matter for health or welfare, and how they compare to the cost of congestion relief, are questions we leave to future work.

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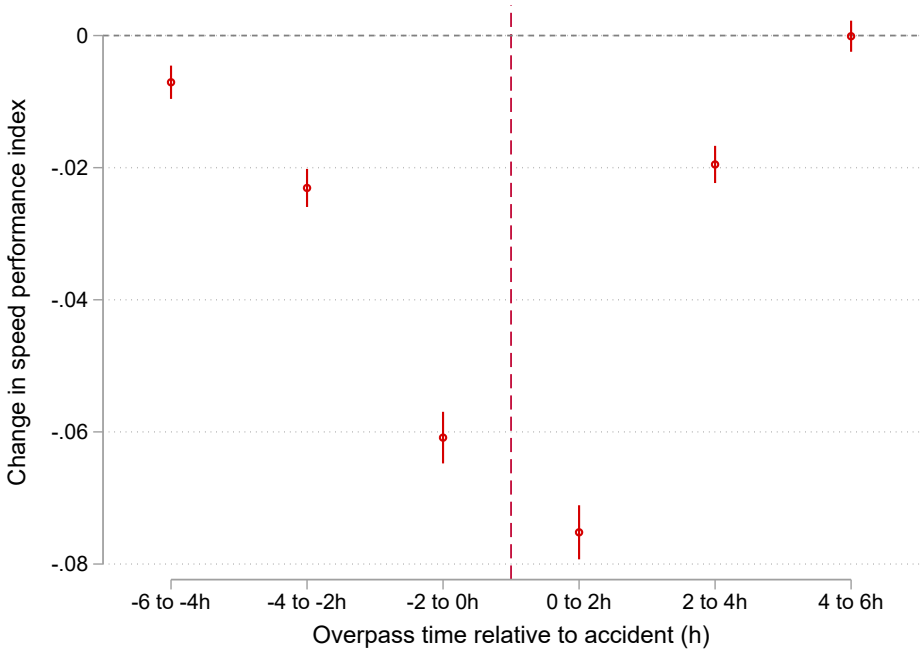
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Appendix Additional Figures and Tables

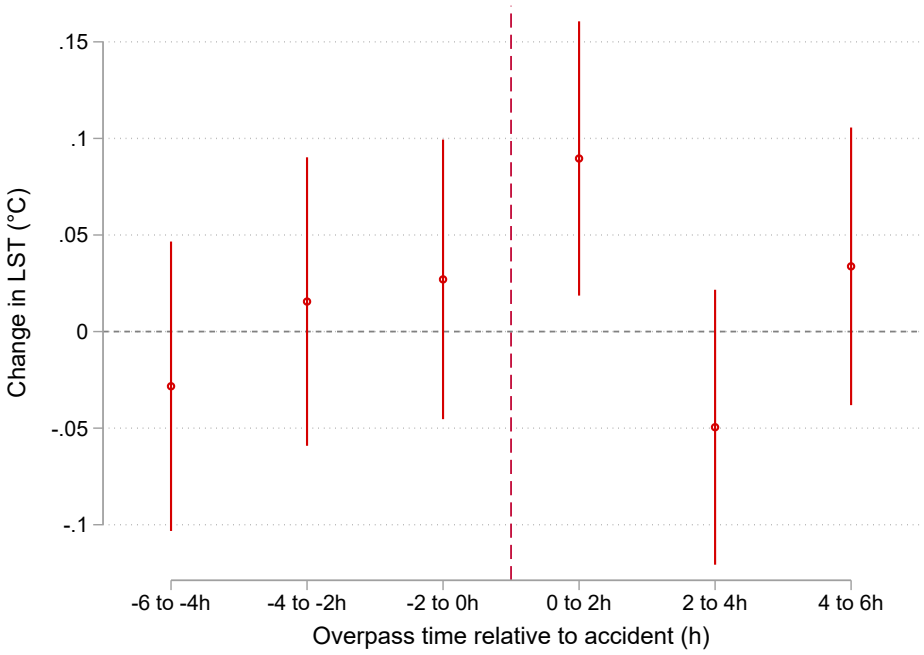
Figures

Figure A.1: Event study, all accidents: speed performance index by time relative to the accident



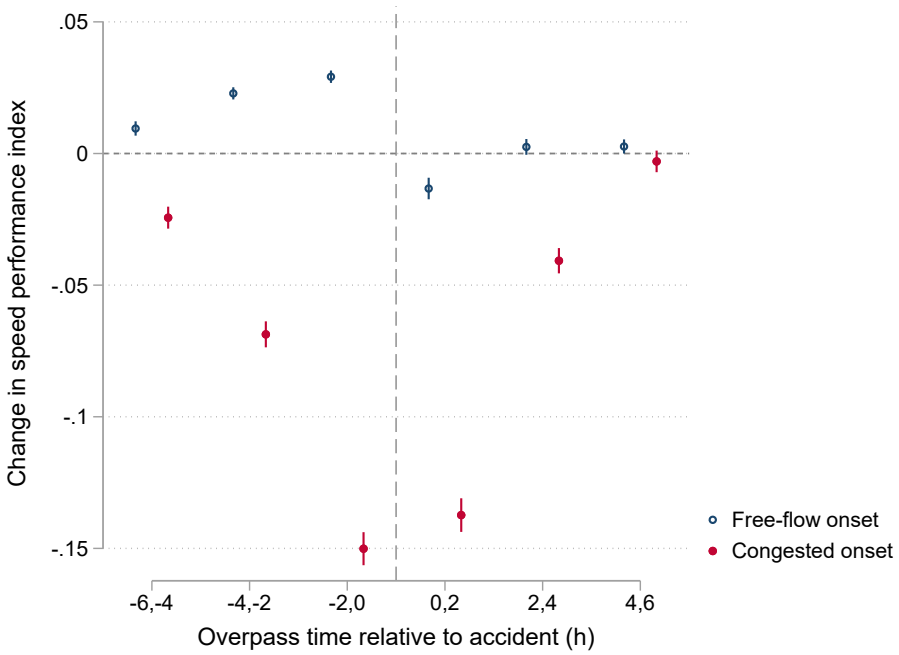
Notes: Estimated $\hat{\phi}_{\ell,u}$ from equation (1) without the weather controls, $Y = \text{SPI}$. Detector + date + hour-of-day fixed effects; 95% CIs clustered at the detector; baseline = no accident within ±6 hours.

Figure A.2: Event study, all accidents: surface temperature by time relative to the accident



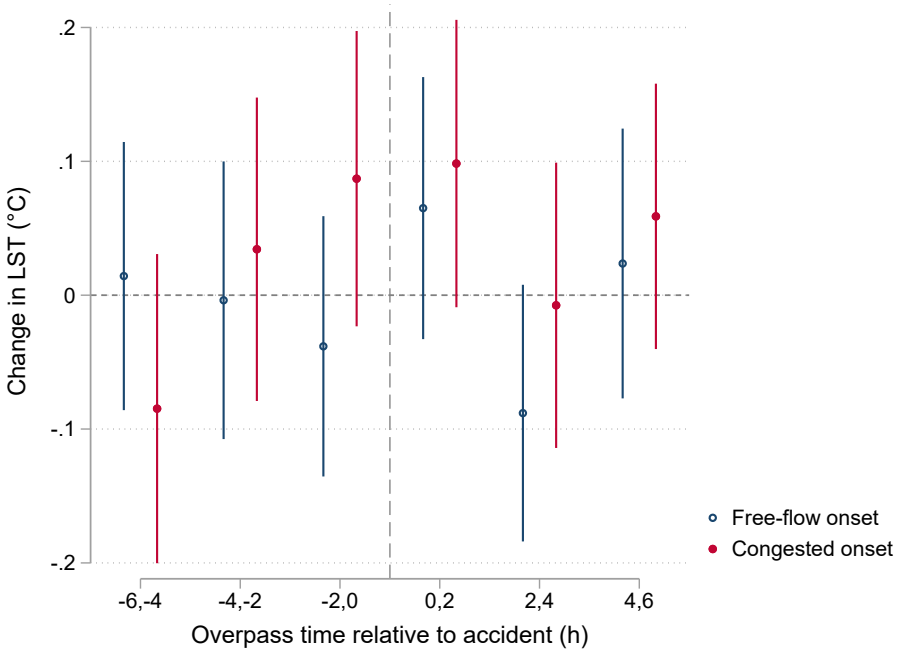
Notes: Same specification as Figure A.1, $Y = \text{LST}$ (°C) in the 500-meter buffer.

Figure A.3: Event study by accident onset, 500-meter buffer, without weather controls: speed performance index



Notes: Free-flow-onset accidents (segment free-flowing in the pre-accident hour) versus congested-onset accidents, estimated jointly. Detector + date + hour-of-day FE; 95% CIs clustered at the detector; baseline = no accident within ±6 hours.

Figure A.4: Event study by accident onset, 500-meter buffer, without weather controls: surface temperature



Notes: Same specification as Figure A.3, $Y = \text{LST}$ (°C).

Tables

Table A.1: First stage: freeway accidents and the speed performance index

	(1) All accidents	(2) Free-flow onset	(3) Congested onset
Accident, any onset (prior 2h)	-0.0765***		
	(0.00210)		
Free-flow-onset accident (prior 2h)		-0.0141***	
		(0.00211)	
Congested-onset accident (prior 2h)			-0.142***
			(0.00328)
First-stage F (cluster-robust)	1,322.4	44.9	1,870.8
Treated (instrument=1)	10,075	5,096	4,950
Observations	3,623,586	3,618,607	3,618,461

Notes: Dependent variable is the speed performance index (speed / free-flow speed; lower = more congested), without weather controls. Detector (α_i), date (μ_d), and hour-of-day (λ_h) fixed effects; SE clustered at the detector. "All" uses every accident; "free-flow onset" restricts to accidents that struck while the segment was free-flowing (pre-accident SPI $\geq$ 0.90); "congested onset" restricts to accidents during existing congestion. Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.2: Effect of congestion on land surface temperature: OLS and 2SLS

	(1) OLS	(2) 2SLS (all)	(3) 2SLS (free-flow)	(4) 2SLS (+vol)
Speed performance index	0.0988*	-1.172**	-4.631	-1.218**
	(0.0585)	(0.475)	(3.586)	(0.481)
Traffic flow (1,000 veh/hr)				-0.0701***
				(0.0117)
Treated (instrument=1)	10,075	10,075	5,096	10,075
Observations	3,623,586	3,623,586	3,618,607	3,623,586

Notes: Outcome is LST (°C) in the 500-meter buffer, without weather controls; $\alpha_i + \mu_d + \lambda_h$ fixed effects; SE clustered at the detector. Columns: OLS; 2SLS with all accidents; 2SLS with free-flow-onset accidents (binary instrument); and 2SLS with all accidents adding contemporaneous traffic volume. Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: 2SLS estimates by instrument and with spatial demeaning

	(1) Binary	(2) Duration	(3) Overidentified	(4) Spatially demeaned
Speed performance index	-7.409**	-7.120**	-7.646***	-3.528
	(3.195)	(3.295)	(2.866)	(2.686)
Reduced form (°C)	0.1143**	–	–	0.0187
	(0.0467)			(0.0128)
KP first-stage F	47.8	33.6	28.0	9.0
Hansen J p (overid)	–	–	0.736	–
Free-flow-onset treated	4,642	4,495	4,495	4,492
Observations	3,304,407	3,304,260	3,304,260	3,224,951

Notes: 2SLS estimates of the effect of the speed performance index on LST (°C) in the near-road ring, with weather controls. The instrument is the free-flow-onset accident indicator in column (1), its duration in column (2), and both (overidentified) in column (3). Column (4) uses the binary instrument with both the outcome and the speed performance index demeaned against detectors in the same three-kilometer cell observed at the same overpass. The reduced form regresses the column's outcome on the free-flow-onset indicator. Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Overidentified 2SLS estimates by disk radius, with and without weather controls

	100 m disk		250 m disk		500 m disk	
	(1) No weather	(2) + Weather	(3) No weather	(4) + Weather	(5) No weather	(6) + Weather
Speed performance index	-6.639**	-7.646***	-6.211**	-7.200**	-5.832*	-6.672**
	(3.050)	(2.866)	(3.027)	(2.866)	(3.054)	(2.887)
KP first-stage F	27.4	28.0	28.0	28.3	27.5	27.8
Hansen J p (overid)	0.711	0.736	0.883	0.819	0.866	0.761
Free-flow-onset treated	4,711	4,495	4,848	4,625	4,942	4,713
Observations	3,454,164	3,304,260	3,554,856	3,400,574	3,618,453	3,461,346

Notes: 2SLS estimates of the effect of the speed performance index on LST (°C) in cumulative disks of the stated radius, instrumented by the free-flow-onset accident indicator and its duration (overidentified). "+ Weather" columns add relative humidity, wind speed, and precipitation at the detector and overpass hour. Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Reduced form without weather controls, by distance ring

	(1) 0–100 m (near-road)	(2) 100–500 m	(3) 500–1000 m
Free-flow-onset accident (prior 2h)	0.0936*	0.0706	0.0335
	(0.0496)	(0.0483)	(0.0494)
Free-flow-onset treated	4,861	4,817	5,036
Observations	3,454,314	3,412,625	3,574,973

Notes: Reduced-form regressions of ring-mean LST (°C) on the free-flow-onset accident indicator, on the sample pairing free-flow-onset-treated overpasses with no-accident controls, without weather controls; Table 3 adds them. All columns include detector, date, and hour-of-day fixed effects. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Prior-day placebo: speed performance index and land surface temperature

	(1) Speed performance index	(2) LST (°C)
Prior-day free-flow-onset accident	0.0090***	-0.0990**
	(0.0016)	(0.0470)
Prior-day treated	4,576	4,576
Observations	3,299,765	3,299,765

Notes: Regressions of the speed performance index (column 1) and LST in the near-road ring (column 2) on an indicator for a free-flow-onset accident at the same time of day on the previous calendar day, estimated with weather controls on the no-accident overpasses with a near-road-ring LST. Detector, date, and hour-of-day fixed effects; standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: 2SLS estimate by spatial clustering unit

	(1) Detector	(2) 5 km grid	(3) 10 km grid	(4) Corridor
Speed performance index	-7.409**	-7.409**	-7.409**	-7.409**
	(3.195)	(3.220)	(3.362)	(3.367)
Clusters	8,850	945	464	194
Observations	3,304,407	3,304,407	3,304,407	3,304,407

Notes: The preferred estimate (binary free-flow-onset instrument, LST in the near-road ring, weather controls, detector, date, and hour-of-day fixed effects) with standard errors clustered at progressively coarser spatial units: the detector (baseline), roughly 5- and 10-kilometer grid cells, and the freeway-by-direction corridor. The point estimate is identical by construction; only the standard error changes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Weather and the free-flow-onset instrument

	(1) Free-flow-onset accident
Air temperature (°C)	2.46**
	(1.20)
Relative humidity (%)	0.63**
	(0.28)
Wind speed (m/s)	-4.86
	(3.01)
Precipitation (mm)	0.77
	(3.06)
Observations	3,461,497
Within R-squared	0.00002

Notes: Regression of the free-flow-onset accident indicator on the weather controls, on the sample pairing free-flow-onset-treated overpasses with no-accident controls. Coefficients and standard errors are multiplied by 10^5 for readability; the instrument's base rate is about 1.4×10^{-3}; the joint test of the four controls gives $F = 2.63$ ($p = 0.03$). Detector, date, and hour-of-day fixed effects; standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9: 2SLS and reduced-form estimates under alternative specifications

	A. Wind direction		B. Lookback window	
	Downwind	Upwind	$H = 1$ hour	$H = 3$ hours
Speed performance index	-5.558*	-6.318*	-3.012	-6.696*
	(3.235)	(3.292)	(3.327)	(4.038)
Reduced form (°C)	0.0837*	0.0949**	0.0619	0.0648*
	(0.0473)	(0.0475)	(0.0679)	(0.0374)
KP first-stage F	47.6	47.2	43.9	32.7
Free-flow-onset treated	4,829	4,826	2,371	6,983
Observations	3,434,911	3,434,456	3,302,136	3,304,419

	C. Adjacent segments		D. Data completeness
	Baseline	Drop ±2 neighbors	
Speed performance index	-7.409**	-9.077***	-6.919**
	(3.195)	(3.522)	(2.946)
Reduced form (°C)	0.1143**	0.1325***	0.1291**
	(0.0467)	(0.0475)	(0.0523)
KP first-stage F	47.8	41.1	55.0
Free-flow-onset treated	4,642	4,499	3,862
Observations	3,304,407	3,292,724	2,797,818

	E. Nearest pixel	F. Maximum LST
Speed performance index	-6.631**	-6.625**
	(3.359)	(3.377)
Reduced form (°C)	0.0986**	0.0989**
	(0.0478)	(0.0487)
KP first-stage F	42.1	47.2
Free-flow-onset treated	4,384	4,864
Observations	3,109,520	3,461,478

Notes: 2SLS estimates of the effect of the speed performance index on LST (°C), and the reduced form of LST on the free-flow-onset indicator, with the binary free-flow-onset instrument and weather controls. Outcome: LST in the near-road ring unless stated. Panel A: mean LST in the downwind and upwind halves of the 500-meter buffer, split by ERA5-Land wind direction at the overpass hour. Panel B: the instrument defined over a one- or a three-hour lookback window before the overpass (baseline: two hours). Panel C: the second column drops control observations within two detector stations of an accident. Panel D: detector-hours with at least 90 percent of five-minute intervals observed (baseline: 50 percent). Panel E: LST of the single ECOSTRESS pixel nearest the detector. Panel F: maximum quality-passing pixel LST in the 500-meter buffer. Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.10: Vehicle-composition bound

	Measured	Selection-adjusted
Change in the speed performance index	-0.0154	-0.036
Added vehicles per kilometer	10.8	25.3
Added vehicles in the 200-meter segment	2.2	5.1
Added vehicle area (m^2)	22	51
Share of the near-road ring's area (%)	0.07	0.16
Implied rise in pixel-mean LST (°C)	0.014	0.032
Reduced form (°C)	0.114	0.151
Bound as a share of the reduced form (%)	12	21
Reduced form net of the bound (°C)	0.101	0.119

Notes: Density rises with the fall in the speed performance index ($k/k_{\text{jam}} = 1 - \text{SPI}$), at a jam density of 140 vehicles per kilometer per lane over five lanes. About 200 meters of freeway cross the near-road ring (radius 100 meters, about 31,000 m^2, 70-meter pixels). Each vehicle covers about 10 m^2, and vehicle surfaces are taken to be 20°C hotter than their surroundings. The selection-adjusted column nets the pseudo-event placebo's first stage (+0.0207) and reduced form (−0.037; Table 7) out of the measured ones. Rows are computed from unrounded inputs.

Table A.11: Reduced form and first stage by air temperature at the overpass

	Quartile of air temperature at the overpass (°C)			
	(1)	(2)	(3)	(4)
	Below 14.5	14.5 to 18.8	18.8 to 24.1	Above 24.1
A. Reduced form: LST in the near-road ring (°C)				
Free-flow-onset accident	0.1003	0.0702	0.0941	0.0587
	(0.0767)	(0.0714)	(0.0857)	(0.0984)
B. First stage: speed performance index				
Free-flow-onset accident	-0.0240***	-0.0216***	-0.0129***	-0.0097**
	(0.0040)	(0.0041)	(0.0039)	(0.0048)
Mean air temperature (°C)	11.1	16.7	21.3	29.1
Treated overpasses	1,160	1,211	1,156	1,115
Observations	826,050	826,055	826,016	826,036

Notes: Each column re-estimates the reduced form (panel A) and the first stage (panel B) on the overpasses in one quartile of the ERA5-Land air temperature at the overpass hour, the quartiles being those of the estimation sample. Outcomes are LST in the near-road ring (0–100 meters, °C) and the speed performance index; the regressor is an indicator for a free-flow-onset accident in the prior two hours. Weather controls; detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.